ICM Model

Interpreting ΔCost (del_u), Carbon Position (N), and Firm Outcomes
Equations + worked numerical examples

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Contents

What questions are we answering?

- At a given carbon price Pc, which firms become net buyers or net sellers of credits (and by how much)?
- How do technology adoption and carbon trading translate into a change in unit cost of production (ΔCost = del_u)?
- How does a change in unit cost feed back into firm output (qst) in the model's reduced-form supply response?

Key variables and units (as implemented in code)

Symbol / code name	Meaning	Typical unit	Notes
q0	Baseline output	unit product	e.g., t product/yr or any consistent unit
qst	Output after policy response	unit product	Computed from q0 using elasticity rule
e_b (BASELINE_EI)	Baseline emissions intensity	tCO_2 / unit	Firm-specific
e_t (TARGET_EI)	Target emissions intensity	tCO_2 / unit	Policy parameter
rint = e_b − e_t	Required EI reduction to meet target	tCO_2 / unit	If negative, firm is below target
Pc	Carbon price	\$ / tCO_2	Model searches Pc to clear market
(c_j, Δe_j)	Tech option j: abatement cost and EI reduction	(\$/tCO_2, tCO_2/unit)	From firm MACC sheet
AJ	Achieved EI reduction from adopted tech	tCO_2 / unit	Sum of Δe_j for options with c_j ≤ Pc
tc (unit_tech_cost)	Unit technology cost from adopted tech	\$ / unit	Sum of c_j·Δe_j
unit_carbon_cost_change	Unit carbon settlement term	\$ / unit	Pc·(rint − AJ); negative means revenue
del_u (ΔCost)	Total unit cost change	\$ / unit	del_u = unit_carbon_cost_change + tc
N	Net carbon position	tCO_2	N = Emissions − Allowance; N>0 buyer, N<0 seller
py	Unit product price (u_baseline)	\$ / unit	Must be in same currency as Pc and costs
epsilon	Output elasticity parameter	dimensionless	Controls output response to unit cost change

Model equations (exactly matching the code)

1) Required intensity reduction

The firm must reduce its emissions intensity from baseline to target:

```
rint = e_b − e_t   (tCO₂/unit)
```

2) Technology adoption rule (MACC threshold)

Each firm adopts all technology options with abatement_cost ≤ Pc.

```
Adopt option j if c_j ≤ Pc
```

Achieved EI reduction from adopted tech:

```
AJ = Σ_{j: c_j ≤ Pc} Δe_j   (tCO₂/unit)
```

Unit technology cost (per unit output):

```
tc = Σ_{j: c_j ≤ Pc} (c_j · Δe_j)   ($/unit)
```

3) Unit carbon settlement and total unit cost change (ΔCost)

Remaining EI gap after adoption is (rint − AJ). This determines buyer/seller status and carbon settlement per unit.

```
unit_carbon_cost_change = Pc · (rint − AJ)   ($/unit)
```

Total unit cost change (the variable plotted as ΔCost):

```
del_u = ΔCost = unit_carbon_cost_change + tc
```

4) Output response rule

The model uses a reduced-form output response: unit cost increase reduces output.

```
δ = del_u / py

qst = q0 · (max(1 − δ, 1e-6))^{epsilon}
```

5) Emissions, allowances, and net carbon position

Emissions and allowances scale with qst. (The code caps negative intensities at 0.)

```
Emis = qst · max(e_b − AJ, 0)

Allowance = qst · e_t

N = Emis − Allowance = qst · (rint − AJ)
```

Interpretation:

- $N > 0$ → buyer (needs credits).
- $N < 0$ → seller (surplus credits).

- $N \approx 0$ → close to compliance.

Worked numerical examples (three cases)

All examples use the same model equations above. Units are consistent: Pc and c_j are in $/tCO$_2$; e values in tCO$_2$/unit; tc and del_u in $/unit; N in tCO$_2$.

Example 1 — Buyer with positive ΔCost

Inputs:

Parameter	Value
q0	1000 unit
e_b	1.00 tCO$_2$/unit
e_t	0.80 tCO$_2$/unit
Pc	$18/tCO$_2$
py	$200/unit
epsilon	0.5

Technology options (adopt if c_j ≤ Pc):

Option	c_j ($/tCO$_2$)	Δe_j (tCO$_2$/unit)
1	10	0.05
2	25	0.10
3	5	0.02

Step-by-step results:

- Compute required reduction rint = e_b – e_t.

```
rint = 1.00 − 0.80 = 0.20
```

- Adopt options with c_j ≤ 18 → options 1 and 3.

```
AJ = 0.05 + 0.02 = 0.07

tc = 10·0.05 + 5·0.02 = 0.50 + 0.10 = 0.60

unit_carbon_cost_change = 18·(0.20 − 0.07) = 18·0.13 = 2.34

del_u = 2.34 + 0.60 = 2.94  (positive ΔCost)

δ = del_u/py = 2.94/200 = 0.0147

qst ≈ 1000·(0.9853)^{0.5} ≈ 992.6
```

- Post-tech intensity: e_b – AJ = 1.00 – 0.07 = 0.93.

```
Emis = 992.6·0.93 = 923.1

Allowance = 992.6·0.80 = 794.1

N = Emis − Allowance = 129.0  → buyer (N>0)
```

Example 2 — Seller but still positive ΔCost (tech cost dominates)

Inputs:

Parameter	Value
q0	1000 unit
e_b	1.00 tCO_2/unit
e_t	0.95 tCO_2/unit
Pc	\$18/tCO_2
py	\$200/unit
epsilon	0.5

Technology options (adopt if c_j ≤ Pc):

Option	c_j (\$/tCO_2)	Δe_j (tCO_2/unit)
1	12	0.06
2	8	0.03

Step-by-step results:

```
rint = 1.00 − 0.95 = 0.05
```

- Adopt both options (12 and 8 ≤ 18).

```
AJ = 0.06 + 0.03 = 0.09
```

- Since AJ > rint, the firm over-complies and can be a seller (rint − AJ < 0).

```
tc = 12·0.06 + 8·0.03 = 0.72 + 0.24 = 0.96

unit_carbon_cost_change = 18·(0.05 − 0.09) = 18·(−0.04) = −0.72
(revenue per unit)

del_u = −0.72 + 0.96 = +0.24  (still positive)

δ = 0.24/200 = 0.0012 → qst ≈ 999.4
```

- Post-tech intensity: e_b − AJ = 0.91.

```
Emis ≈ 999.4·0.91 = 909.5

Allowance ≈ 999.4·0.95 = 949.4

N ≈ −39.9  → seller (N<0)
```

Example 3 — Seller with negative ΔCost (credit revenue dominates)

Inputs:

Parameter	Value
q0	1000 unit
e_b	1.00 tCO_2/unit
e_t	0.95 tCO_2/unit
Pc	\$18/tCO_2

py	$200/unit
epsilon	0.5

Technology options (adopt if c_j ≤ Pc):

Option	c_j ($/t$CO_2$)	Δe_j (tCO_2/unit)
1	2	0.08
2	3	0.05

Step-by-step results:

```
rint = 1.00 − 0.95 = 0.05
```

- Adopt both options (2 and 3 ≤ 18).

```
AJ = 0.08 + 0.05 = 0.13
tc = 2·0.08 + 3·0.05 = 0.16 + 0.15 = 0.31
unit_carbon_cost_change = 18·(0.05 − 0.13) = 18·(−0.08) = −1.44
del_u = −1.44 + 0.31 = −1.13  (negative ΔCost; net benefit)
δ = −1.13/200 = −0.00565 → (1−δ)=1.00565 → qst ≈ 1002.8
```

- Post-tech intensity: e_b − AJ = 0.87.

```
Emis ≈ 1002.8·0.87 = 872.4
Allowance ≈ 1002.8·0.95 = 952.7
N ≈ −80.3  → seller (N<0)
```

Why a firm can be a seller but still have positive ΔCost

Seller status depends on (rint − AJ) < 0, but ΔCost includes both carbon settlement and technology cost.

```
Seller if (rint − AJ) < 0  ⇔  AJ > rint
del_u = Pc(rint − AJ) + tc
```

Therefore, a seller can still have del_u > 0 whenever:

```
tc  >  Pc(AJ − rint)
```

- Over-compliance is small (AJ only slightly above rint), so credit revenue per unit is small.
- Technology options adopted under the c_j ≤ Pc rule still have positive costs, so tc can dominate.
- Targets may be generous for some firms; they become sellers even with modest AJ.
- The adoption rule adopts all options below Pc, not only the minimum needed to hit the target.